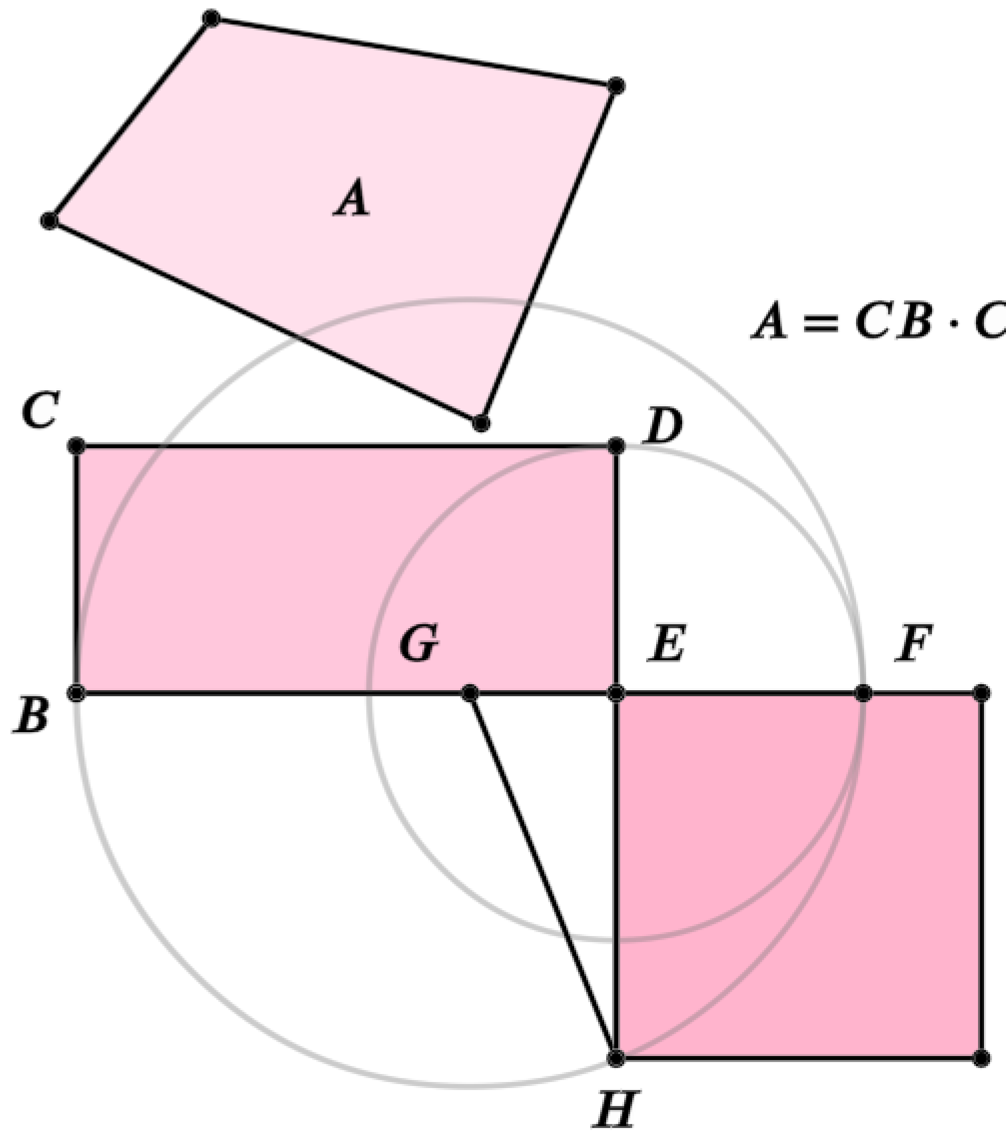
Euclid's Elements

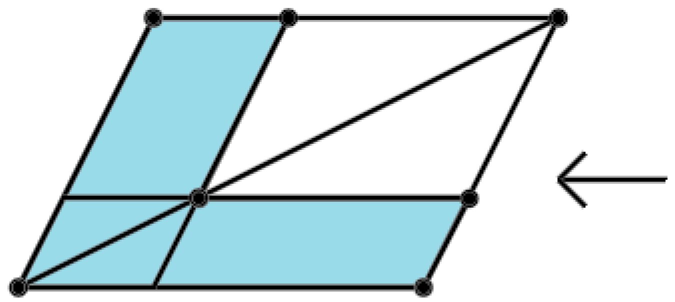
Book II



$$A = CB \cdot CD = (EH)^2$$

It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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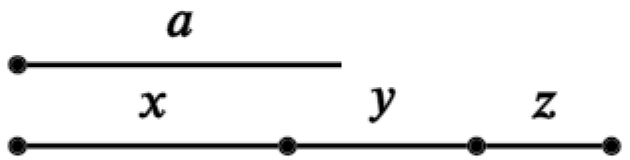


$a\,(x + y + z) = ax + ay + az$

2.

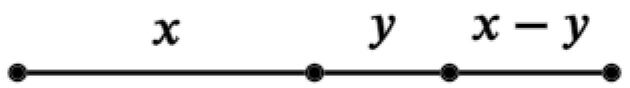

$(x + y)^2 = x\,(x + y) + y\,(x + y)$

3.


$y\,(x + y) = yx + y^2$

4.


$(x + y)^2 = x^2 + y^2 + 2xy$

5.


$(x + y)(x - y) + y^2 = x^2$

6.

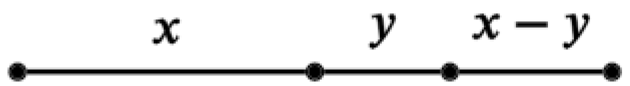

$y\,(2x + y) + x^2 = (x + y)^2$

7.


$(x + y)^2 + y^2 = x^2 + 2y\,(x + y)$

8.


$(x + 2y)^2 = 4y\,(x + y) + x^2$

9.


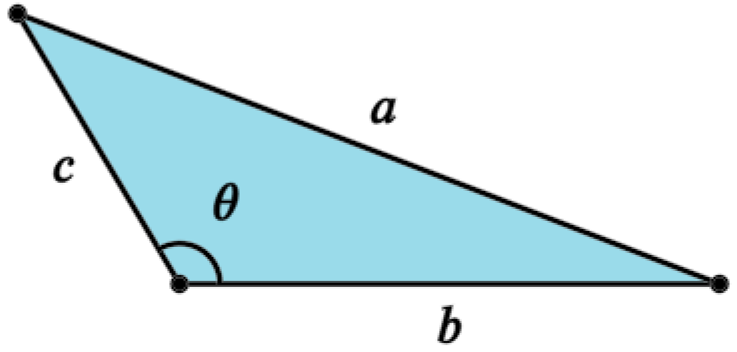
$(x + y)^2 + (x - y)^2 = 2\,(x^2 + y^2)$

10.

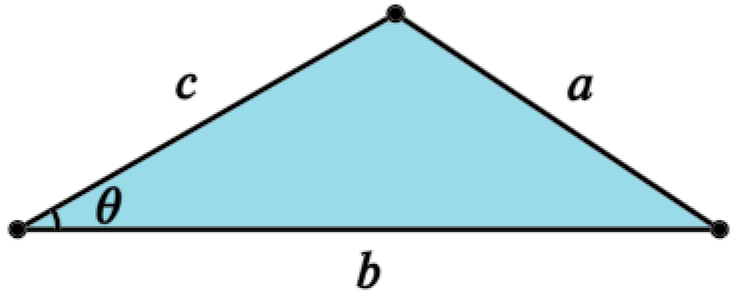

$(2x + y)^2 + y^2 = 2\,(\,x^2 + (x + y)^2\,)$

11.


Find H such that: $y\,(x + y) = x^2$

12.


Cosine Law, $a^2 = b^2 + c^2 - 2bc\cos\theta$

13.


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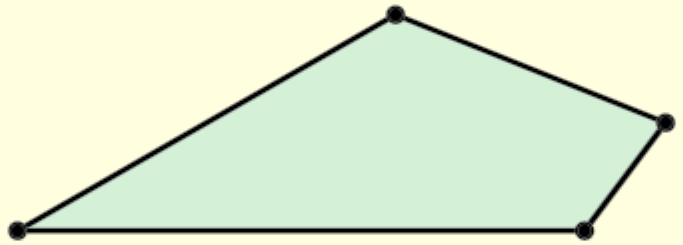
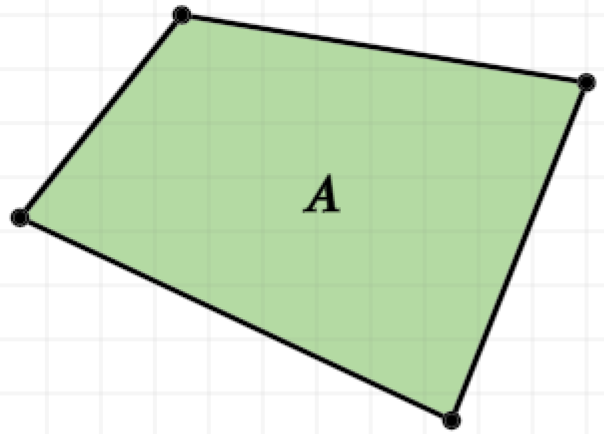


Construct a square equal to the polygon

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.



Construction:

Let A be the given rectilinear figure

Proposition 14 of Book 2

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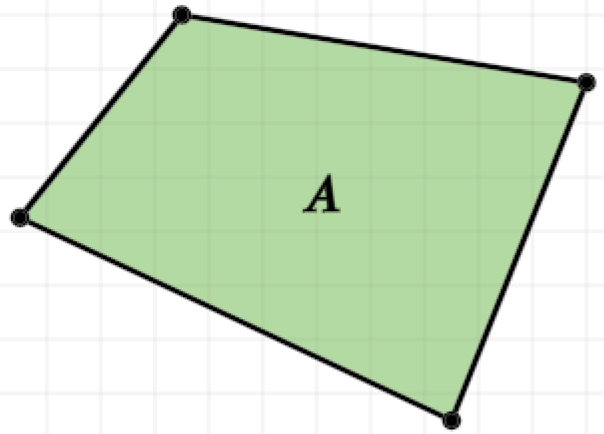
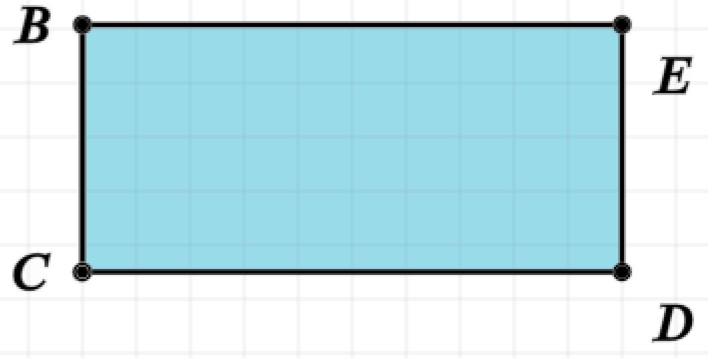


$\square A = \square BD$

Construction:

Let A be the given rectilinear figure

Copy A to a rectangle (I.45)



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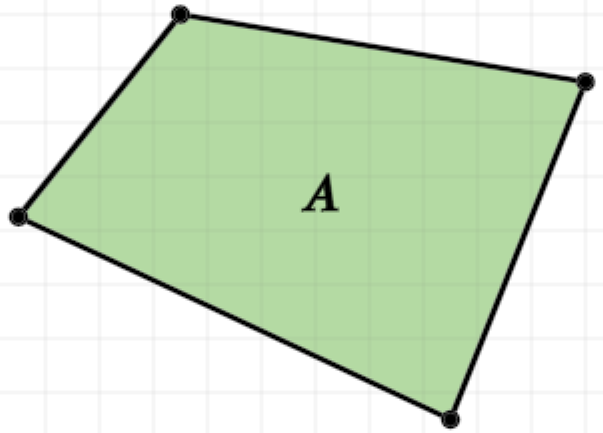


$\blacksquare A = \blacksquare BD$

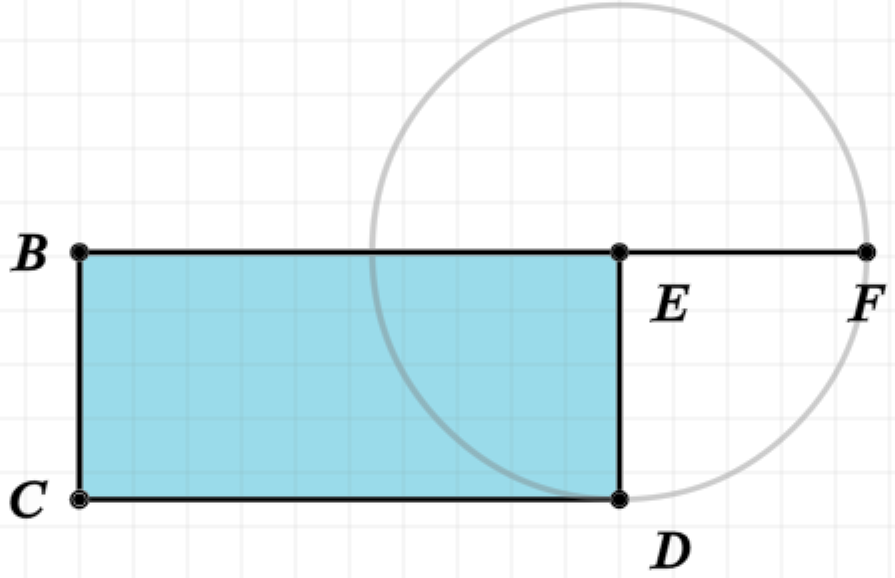
$EF = ED$

Construction:

Let A be the given rectilinear figure

Copy A to a rectangle (I.45)

If BE does not equal ED, and if BE is the larger, extend BE to F, where EF equals ED



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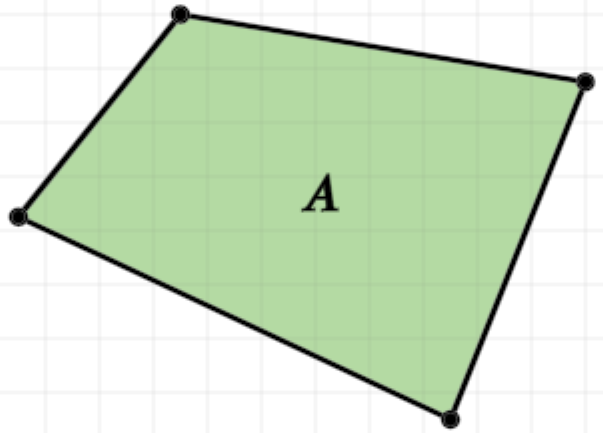


$A = BD$

$EF = ED$

$BG = GF$

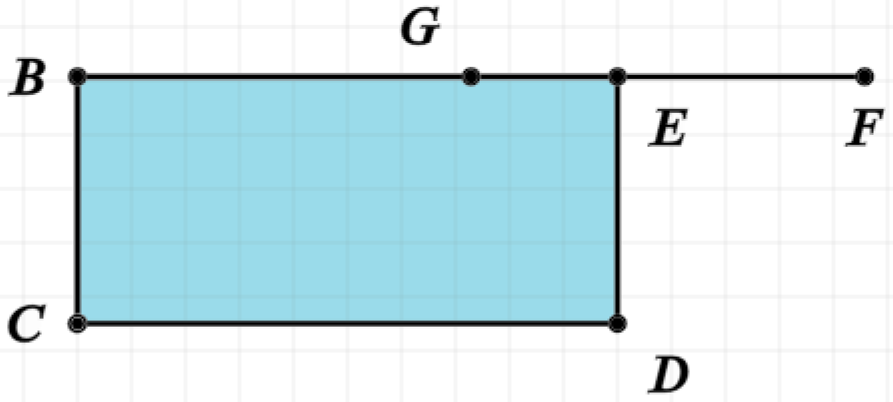
Construction:

Let A be the given rectilinear figure

Copy A to a rectangle (I.45)

If BE does not equal ED, and if BE is the larger, extend BE to F, where EF equals ED

Bisect BF (and label it point G)



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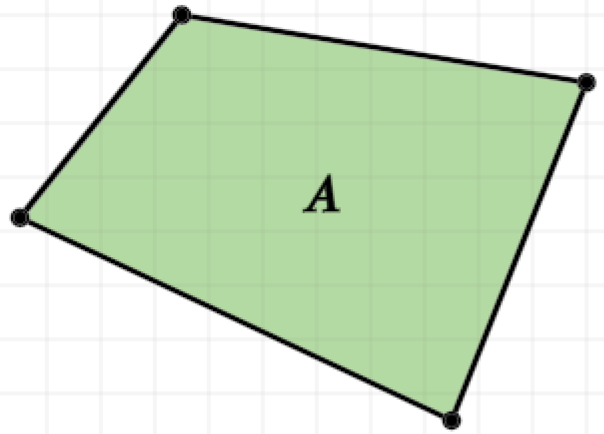

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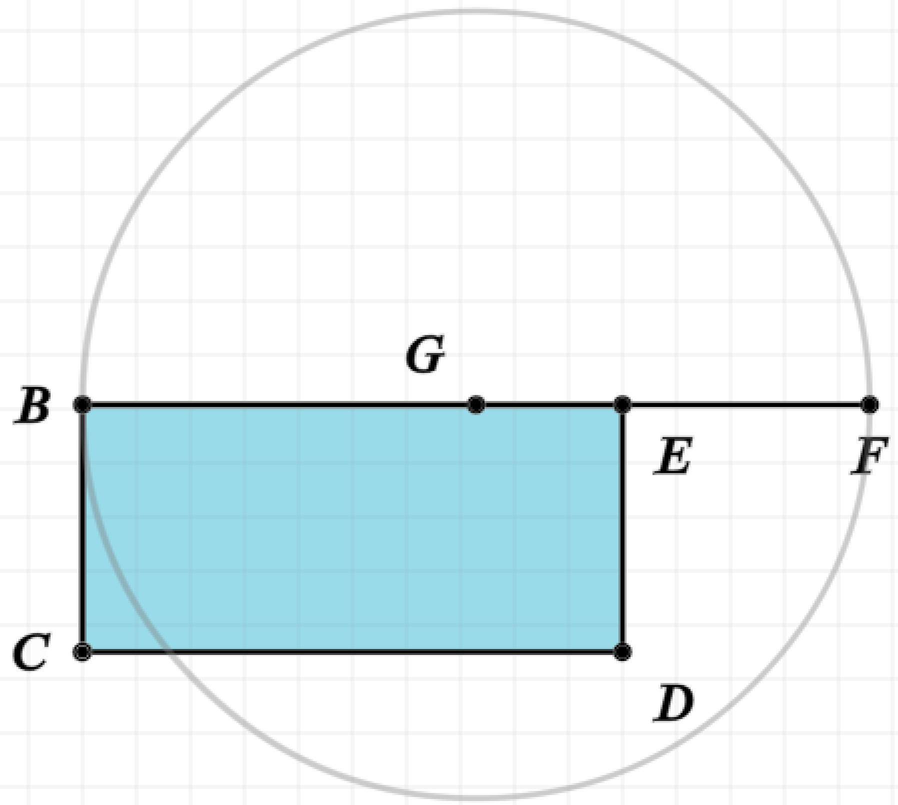

Construction:

Let A be the given rectilinear figure

Copy A to a rectangle (I.45)

If BE does not equal ED, and if BE is the larger, extend BE to F, where EF equals ED

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Draw a circle with G as the center and GF as the radii

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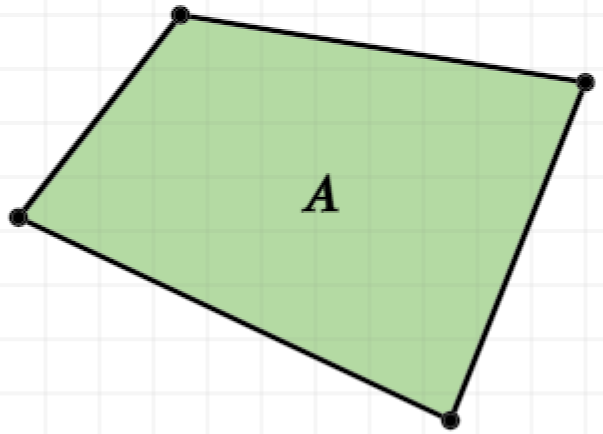
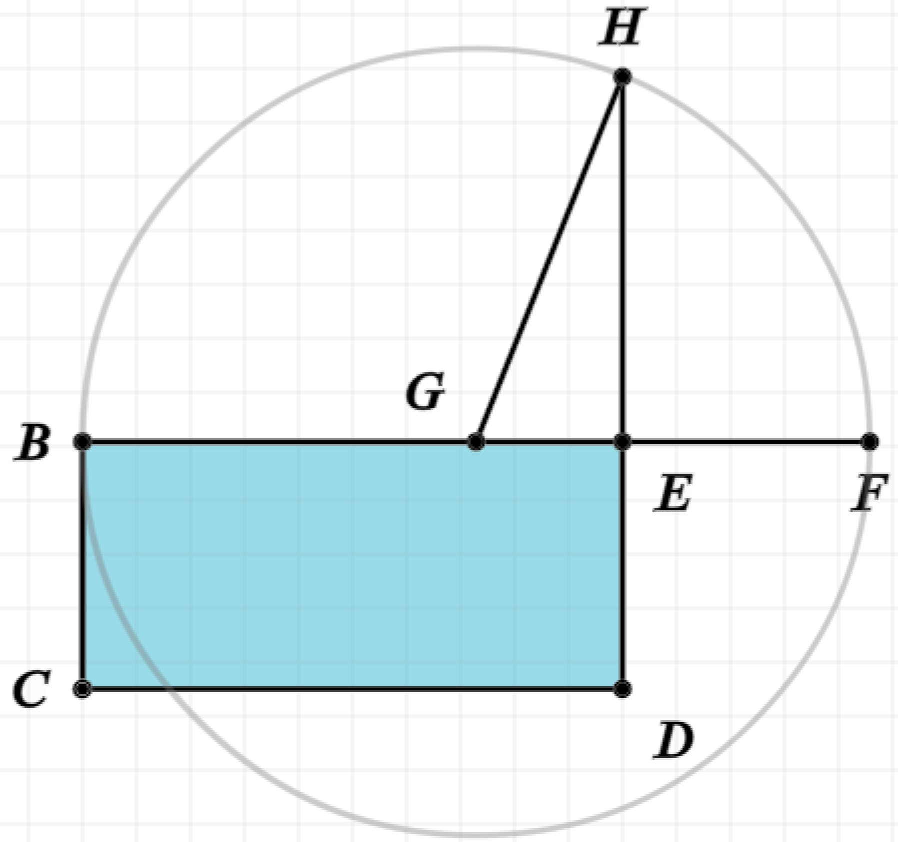


$\square A = \square BD$

$EF = ED$

$BG = GF$

$GH = GF$



Construction:

Let A be the given rectilinear figure

Copy A to a rectangle (I.45)

If BE does not equal ED, and if BE is the larger, extend BE to F, where EF equals ED

Bisect BF (and label it point G)

Draw a circle with G as the center and GF as the radii

Extend DE to intersect with the circle at point H, and let GH be joined

Proposition 14 of Book 2

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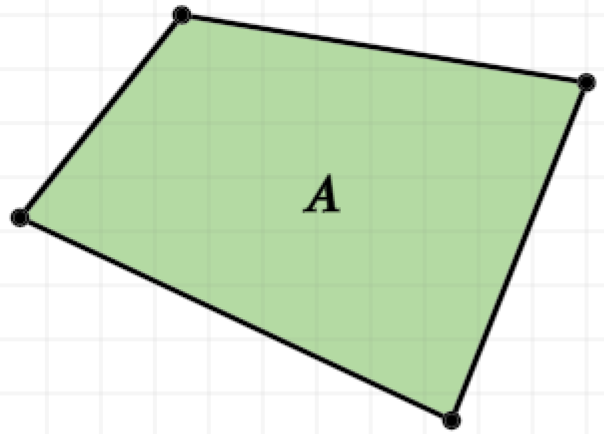


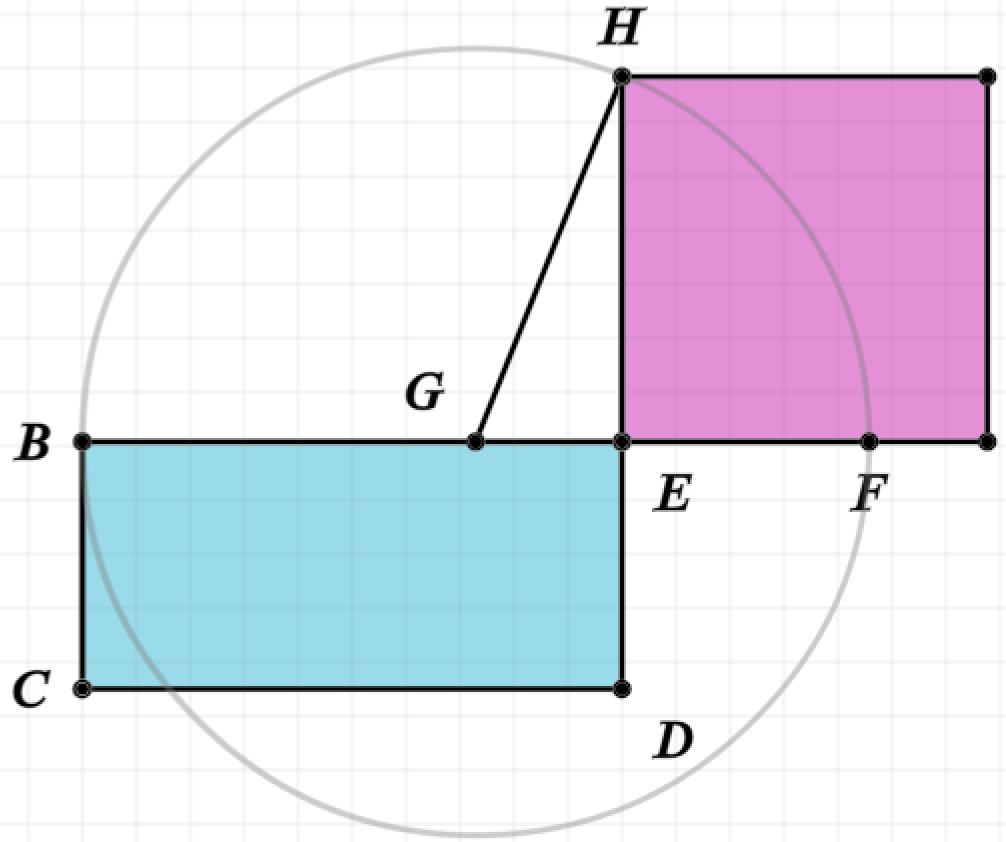
$\square A = \square BD$

$EF = ED$

$BG = GF$

$GH = GF$

$\square HE = \square A$



Construction:

Let A be the given rectilinear figure

Copy A to a rectangle (I.45)

If BE does not equal ED, and if BE is the larger, extend BE to F, where EF equals ED

Bisect BF (and label it point G)

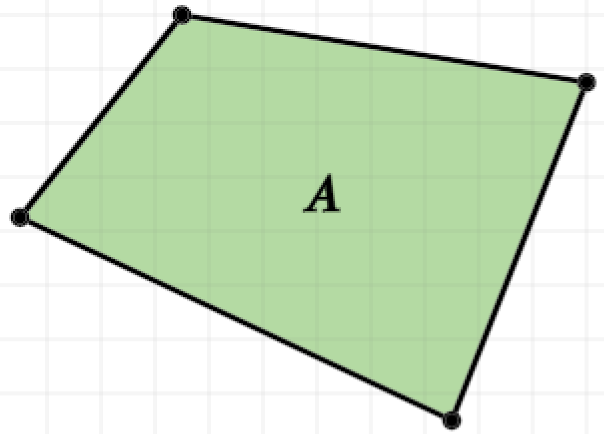
Draw a circle with G as the center and GF as the radii

Extend DE to intersect with the circle at point H, and let GH be joined

The square on HE is equal in area to figure A

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.



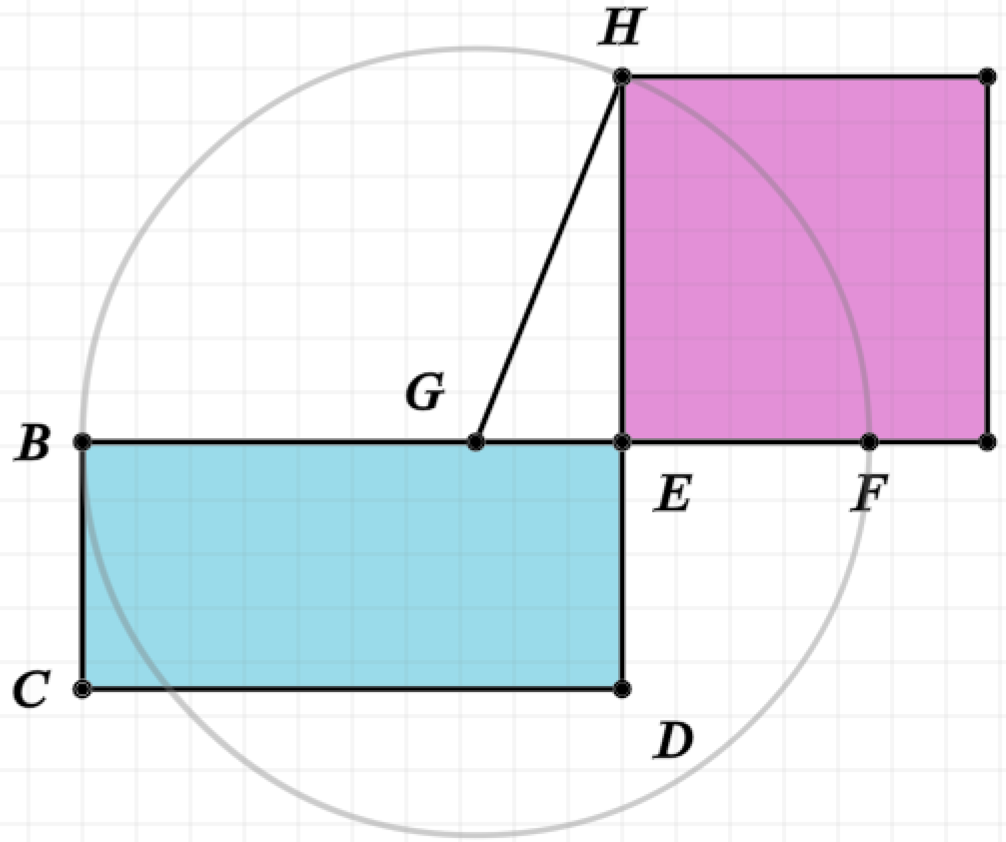
Proof:

$A = BD$

$EF = ED$

$BG = GF$

$GH = GF$



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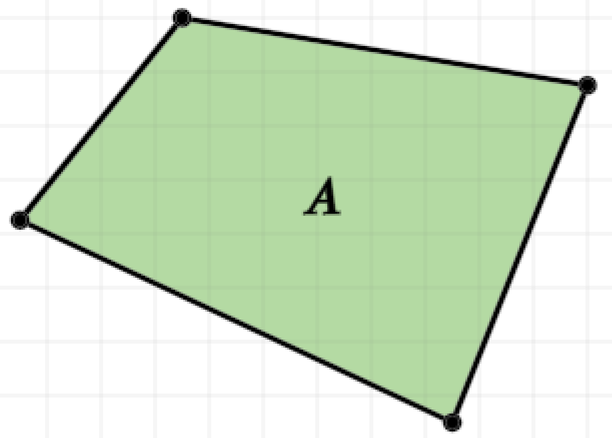

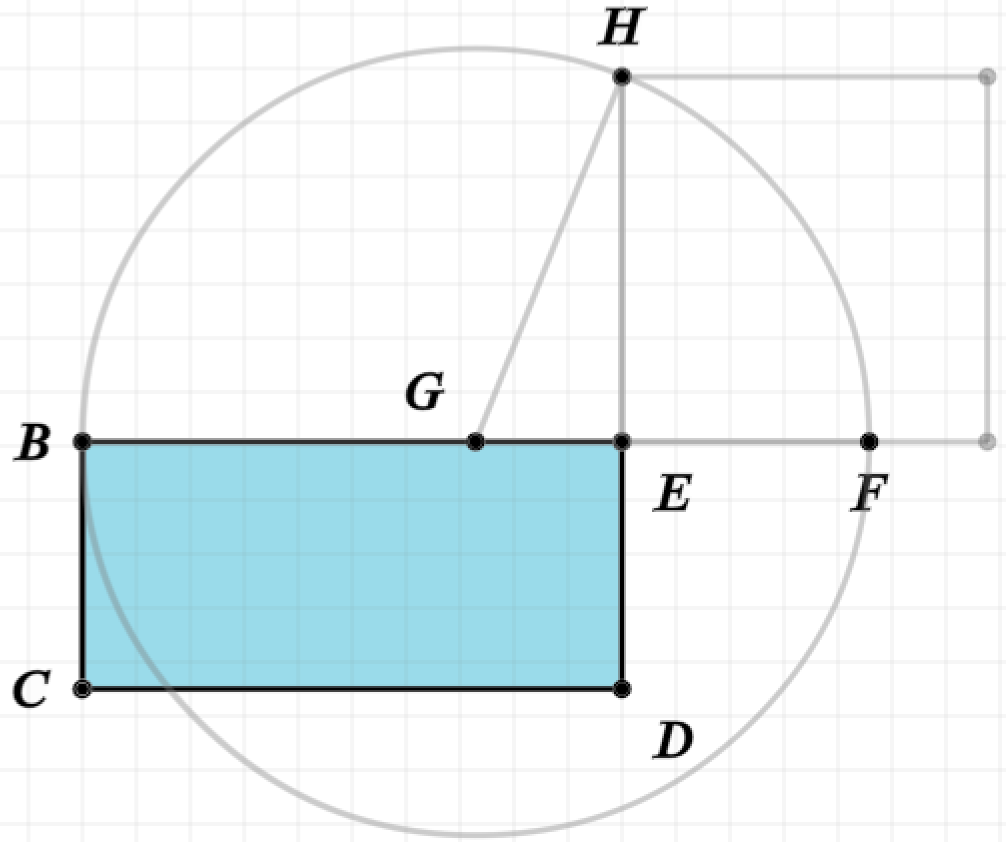

$\square A = \square BD$

$EF = ED$

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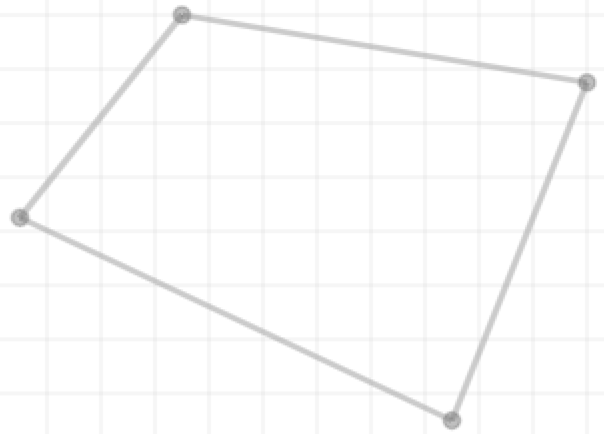
$GH = GF$

Proof:

Polygon A equals the polygon BD by construction

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.



$A = BD$

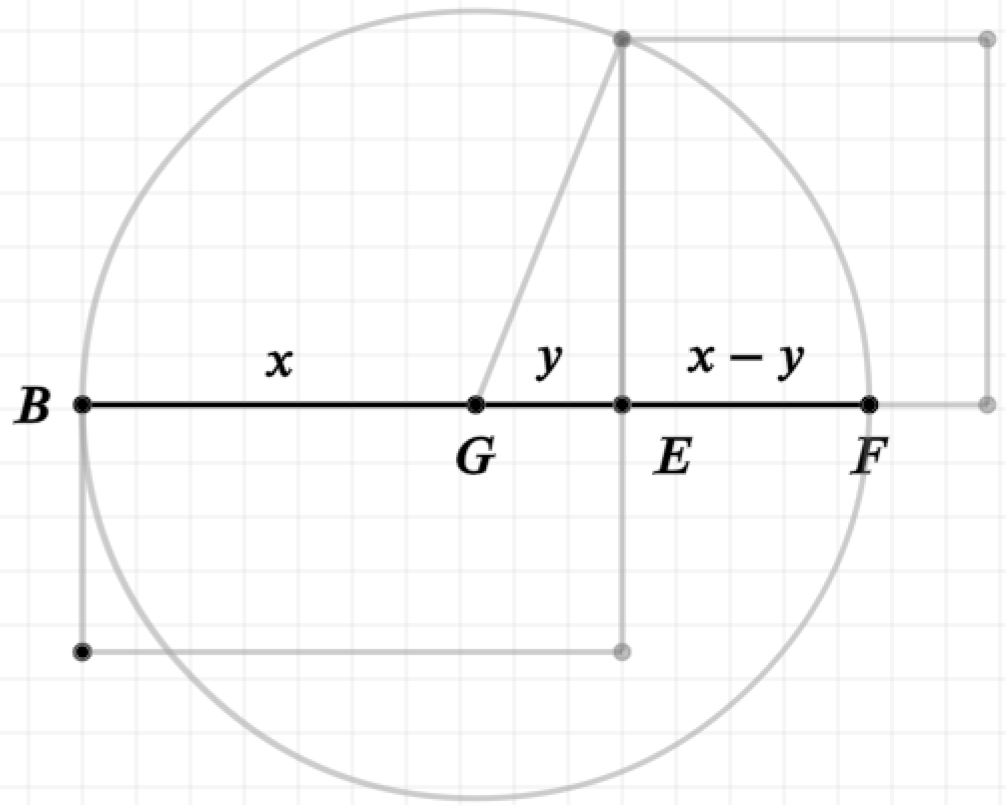
$EF = ED$

$BG = GF$

$GH = GF$

$BE \cdot EF + EG^2 = GF^2$

$(x + y)(x - y) + y^2 = x^2$



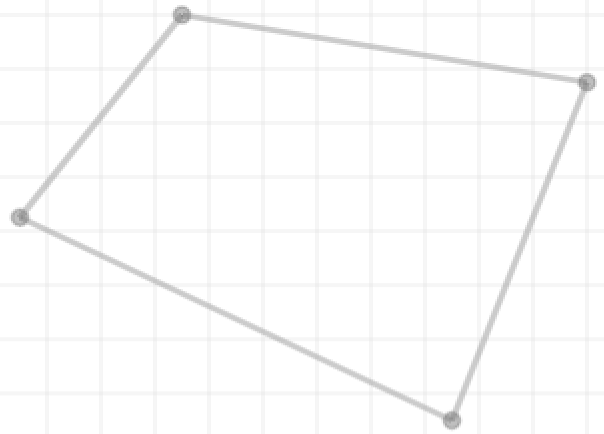
Proof:

Polygon A equals the polygon BD by construction

Line BF is divided into equal (G) and unequal segments (E), thus the rectangle formed by BE,EF plus the square of EG is equal to the square on GF (II.5)

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.



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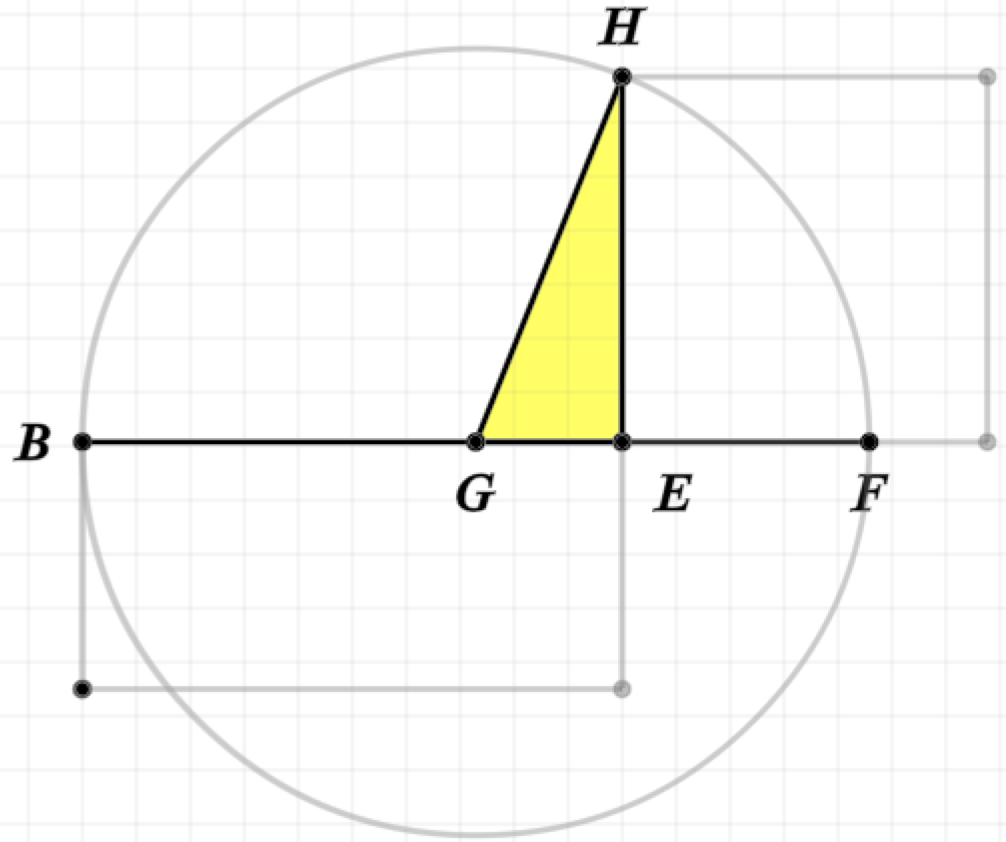


Proof:

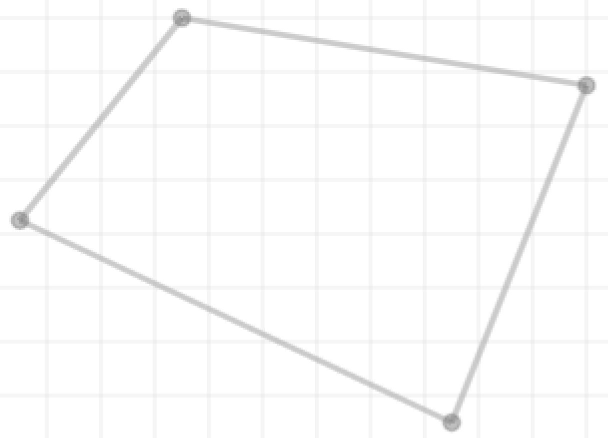
Polygon A equals the polygon BD by construction

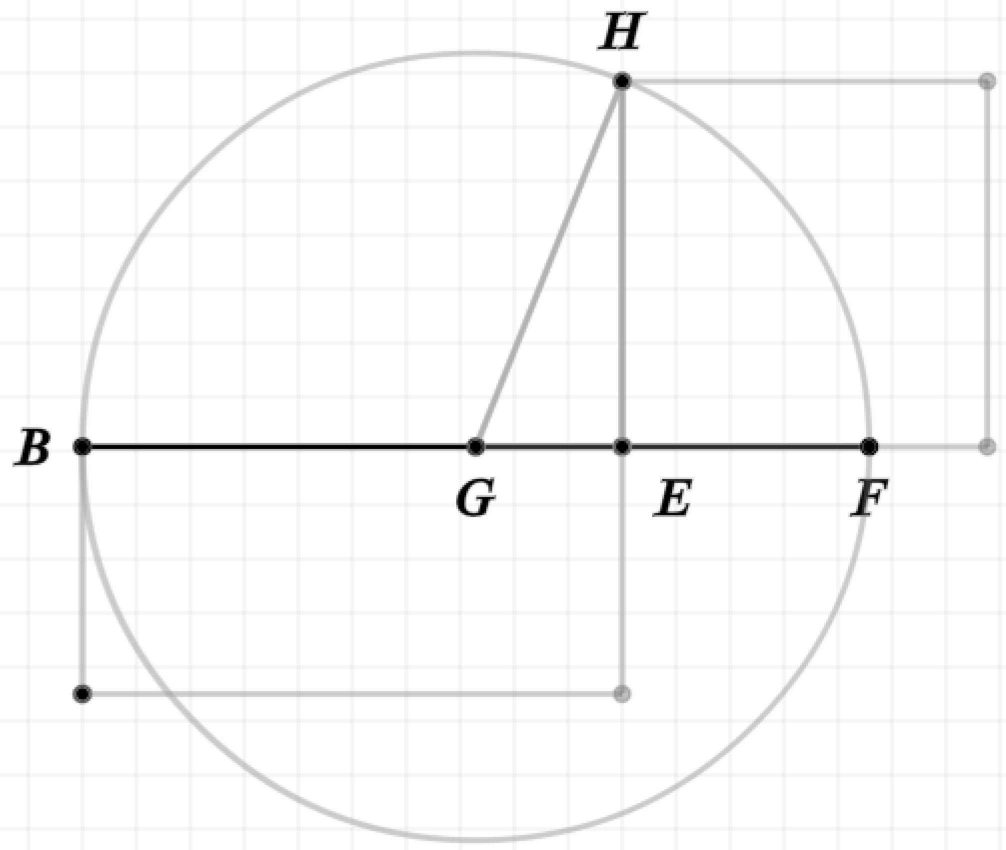
Line BF is divided into equal (G) and unequal segments (E), thus the rectangle formed by BE,EF plus the square of EG is equal to the square on GF (II.5)

Since GHE is a right triangle, and GH is equal to GF, the square of GF is equal to the sum of the squares on EG and GH (I.47)

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.





$A = BD$

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Proof:

Polygon A equals the polygon BD by construction

Line BF is divided into equal (G) and unequal segments (E), thus the rectangle formed by BE,EF plus the square of EG is equal to the square on GF (II.5)

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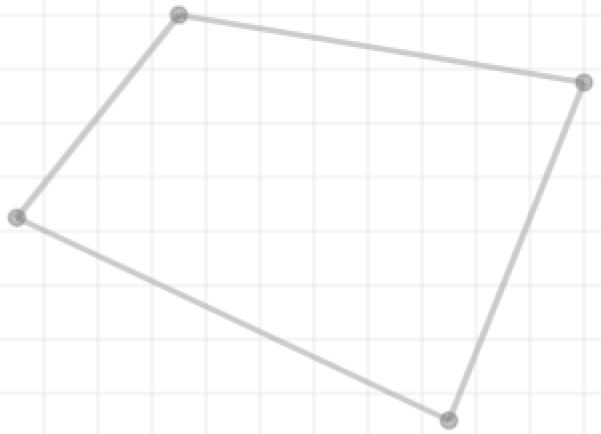

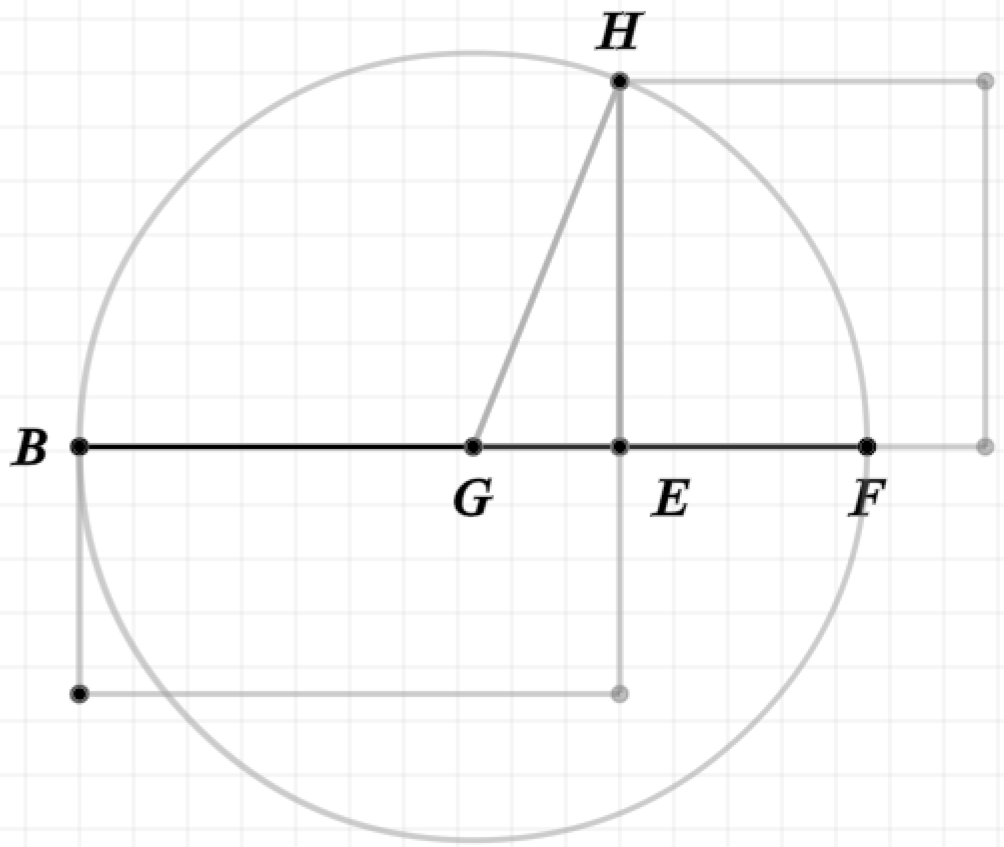

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$$BE \cdot EF = EH^2$$

Proof:

Polygon A equals the polygon BD by construction

Line BF is divided into equal (G) and unequal segments (E), thus the rectangle formed by BE,EF plus the square of EG is equal to the square on GF (II.5)

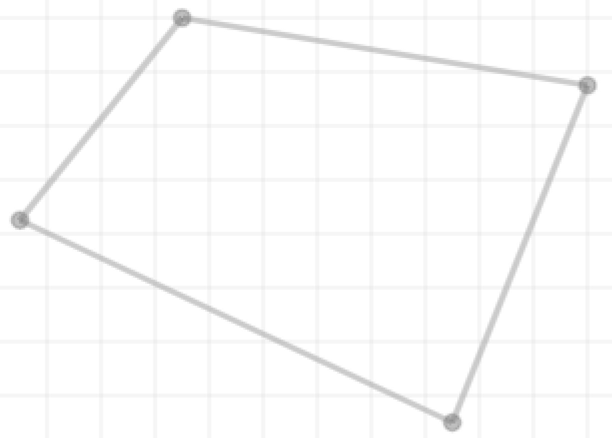
Since GHE is a right triangle, and GH is equal to GF, the square of GF is equal to the sum of the squares on EG and GH (I.47)

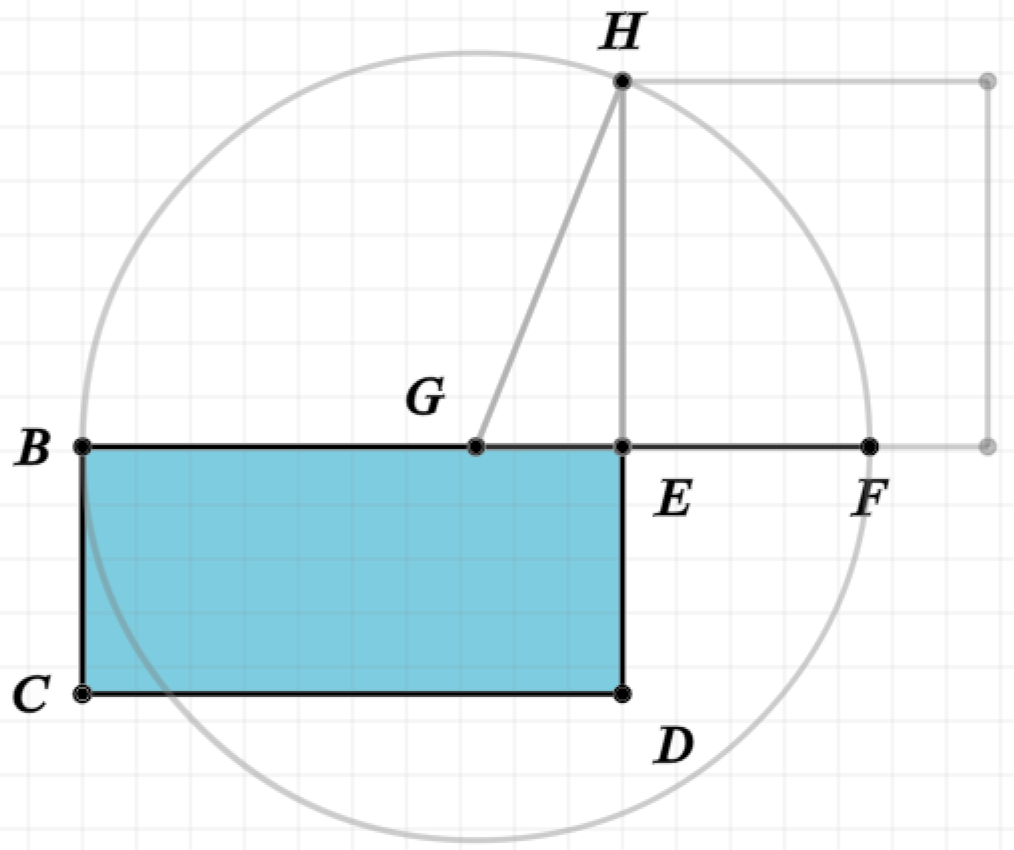
Thus the rectangle formed by BE,EF plus the square of EG is equal to the sum of the squares on EG and GH

Subtracting EG from both sides of the equality, gives BE,EF equals the square of EH

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.





$$A = BD$$

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$$BE \cdot EF + EG^2 = GF^2$$

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$$GF^2 = EG^2 + EH^2$$

$$BE \cdot EF + EG^2 = EG^2 + EH^2$$

$$BE \cdot EF = EH^2$$

$$BE \cdot EF = BD$$

Proof:

Polygon A equals the polygon BD by construction

Line BF is divided into equal (G) and unequal segments (E), thus the rectangle formed by BE,EF plus the square of EG is equal to the square on GF (II.5)

Since GHE is a right triangle, and GH is equal to GF, the square of GF is equal to the sum of the squares on EG and GH (I.47)

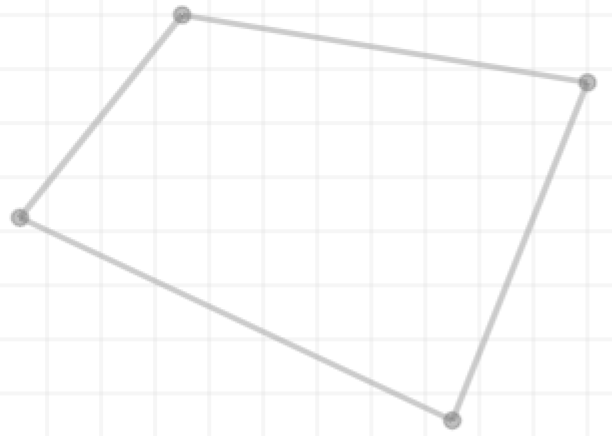
Thus the rectangle formed by BE,EF plus the square of EG is equal to the sum of the squares on EG and GH

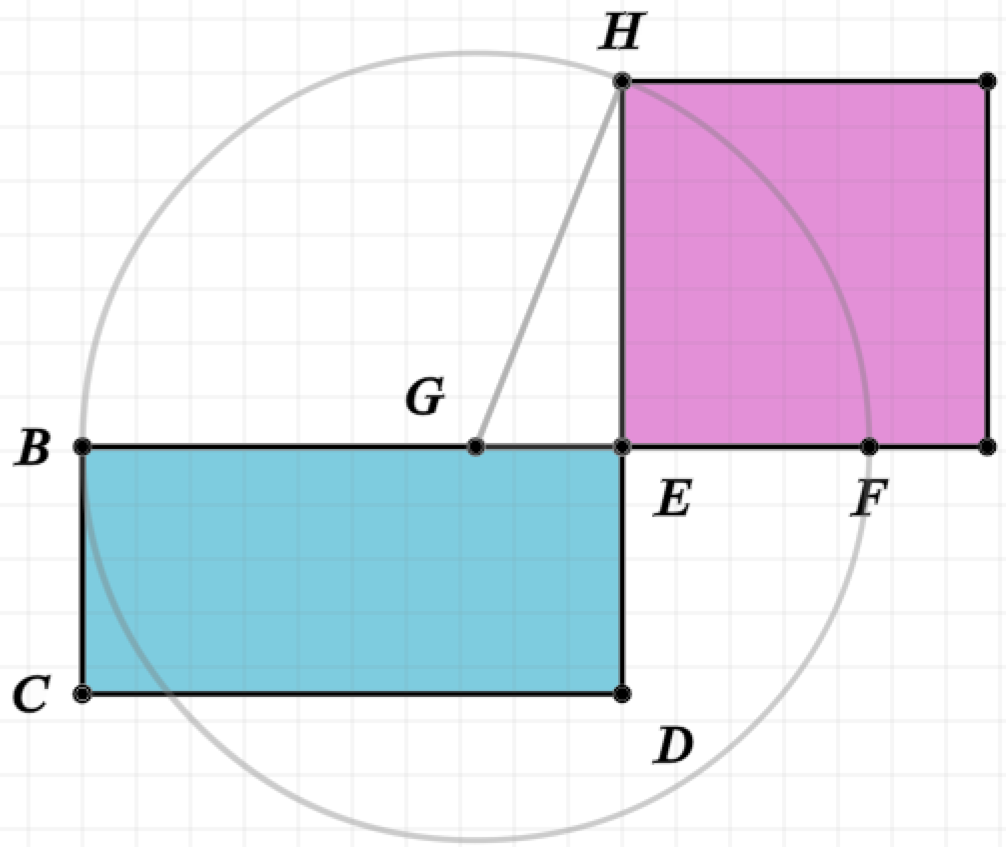
Subtracting EG from both sides of the equality, gives BE,EF equals the square of EH

The rectangle formed by BE,EF is BD, since EF equals ED

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.





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$GF^2 = EG^2 + EH^2$

$BE \cdot EF + EG^2 = EG^2 + EH^2$

$BE \cdot EF = EH^2$

$BE \cdot EF = BD$

$BD = EH^2$

Proof:

Polygon A equals the polygon BD by construction

Line BF is divided into equal (G) and unequal segments (E), thus the rectangle formed by BE,EF plus the square of EG is equal to the square on GF (II.5)

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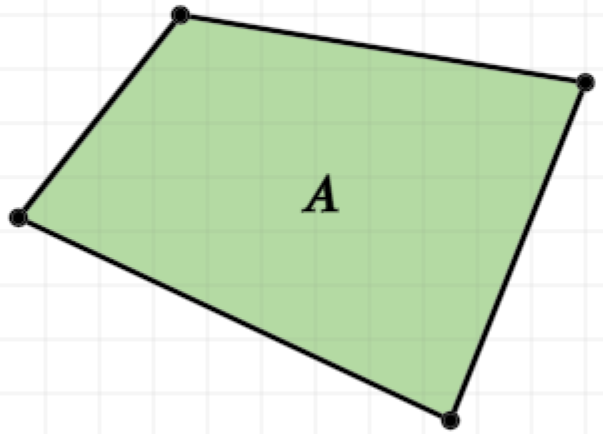
Subtracting EG from both sides of the equality, gives BE,EF equals the square of EH

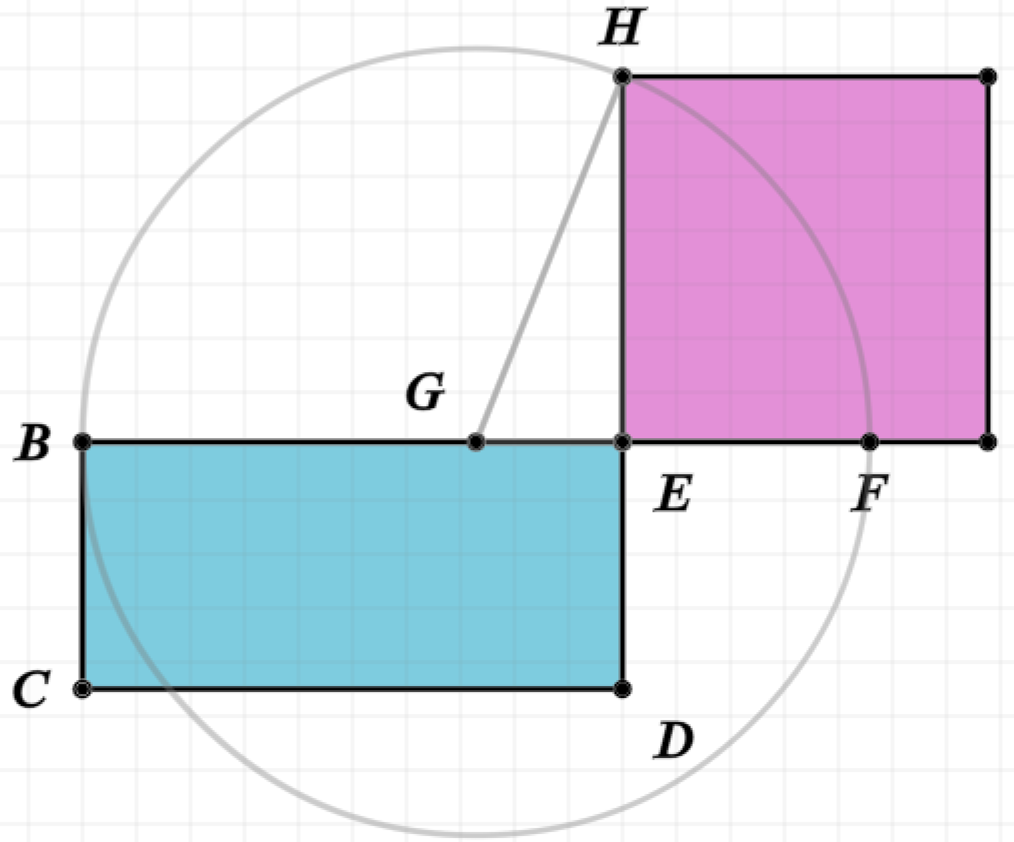
The rectangle formed by BE,EF is BD, since EF equals ED

Therefore BD equals the square on EH

Proposition 14 of Book 2

To construct a square equal to a given rectilinear figure.





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$BE \cdot EF + EG^2 = EG^2 + EH^2$

$BE \cdot EF = EH^2$

$BE \cdot EF = BD$

$BD = EH^2$

$A = EH^2 = EH$

Proof:

Polygon A equals the polygon BD by construction

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Subtracting EG from both sides of the equality, gives BE,EF equals the square of EH

The rectangle formed by BE,EF is BD, since EF equals ED

Therefore BD equals the square on EH

Therefore polygon 'A' equals the square on EH

Proposition 14 of Book 2

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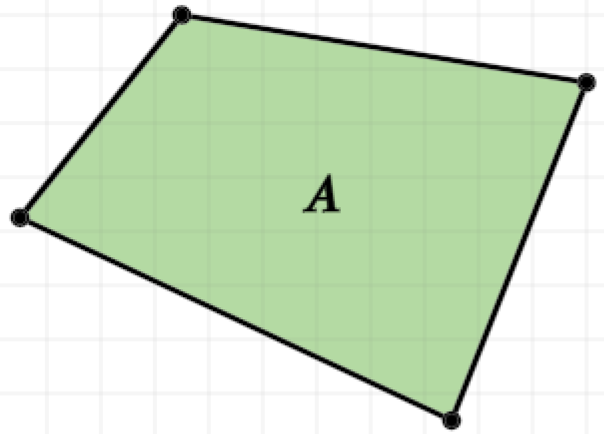

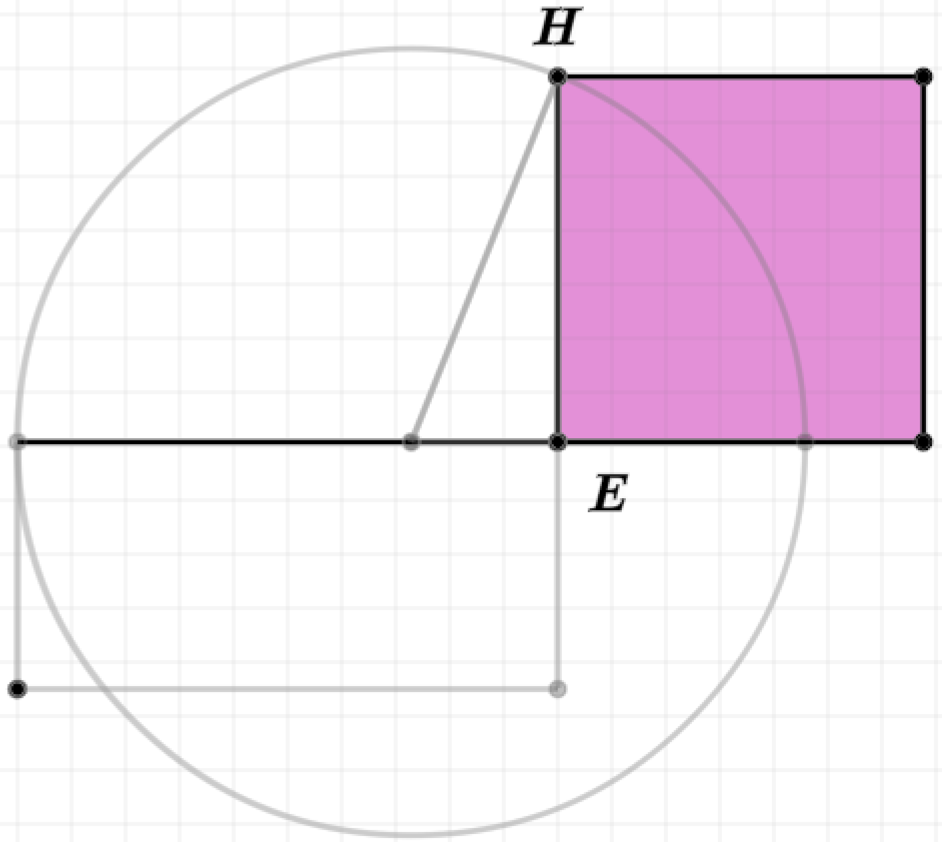

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$BE \cdot EF + EG^2 = EG^2 + EH^2$

$BE \cdot EF = EH^2$

$BE \cdot EF = BD$

$BD = EH^2$

$A = EH^2 = EH$

$A = EH$

Proof:

Polygon A equals the polygon BD by construction

Line BF is divided into equal (G) and unequal segments (E), thus the rectangle formed by BE,EF plus the square of EG is equal to the square on GF (II.5)

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Thus the rectangle formed by BE,EF plus the square of EG is equal to the sum of the squares on EG and GH

Subtracting EG from both sides of the equality, gives BE,EF equals the square of EH

The rectangle formed by BE,EF is BD, since EF equals ED

Therefore BD equals the square on EH

Therefore polygon 'A' equals the square on EH

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